Determine Surface Area of Pyramids

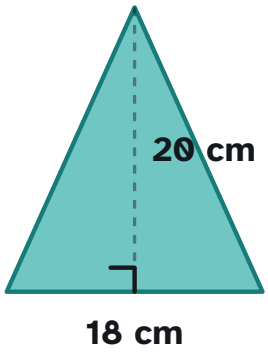
Find the lateral area of a pyramid: only the slanted faces, with the base left out.

Name _____ **Date** _____

Tonight's spark

A model hut is a pyramid with a triangle floor. The floor is 18 cm on each side. Each of the 3 slanted faces has a slant height of 20 cm. You paint only the slanted faces. How many square centimeters get paint?

Guess first: is it more than 500 square centimeters, or less?



Our class model: slanted faces only

Try it

Work it out on paper. Write one answer on each line.

1 **What is the lateral area of a pyramid?**

A. The area of the base only **B.** The area of the slanted triangle faces only
C. The area of all faces including the base **D.** The space inside the pyramid

My answer _____

2 **The triangle faces of a pyramid are 14 square units each, and there are 4 of them. What is the lateral area?**

A. 18 square units **B.** 28 square units **C.** 14 square units **D.** 56 square units

My answer _____

3 **A tent is a square pyramid with 4 triangle sides. Each side has a base of 4 and a slant height of 5. How much cloth is needed for the sides only, not the floor?**

A. 40 square units **B.** 56 square units **C.** 80 square units **D.** 10 square units

My answer _____

4 **A pyramid has a lateral area of 40 and a base area of 16. What is its total surface area?**

A. 24 **B.** 40 **C.** 56 **D.** 640

My answer _____

5 **You want to paint only the sides of a pyramid-shaped hut, not the floor. Which area do you find?**

A. Lateral area **B.** Total surface area **C.** Base area **D.** Volume

My answer _____

Talk about it

Why do we leave the base out when we only paint or cover the slanted sides?

Home mission

Find a gift box and a lampshade or paper cup. For each one, ask: is the bottom covered? Say when you would add the base and when you would leave it out.

If we get stuck, ask

- Is the bottom being covered in this problem?
- How many slanted faces does this pyramid have?
- Are you using the slant height, or the height through the middle?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/5-8-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it